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MONITORING FOR UNSTABLE APPROACH OF AIRCRAFT USING SPECIFIC ENERGY

Abstract

The present disclosure provides a method of monitoring for an unstable approach of an aircraft during flight in one aspect, the method including: determining, using values of a ground speed and a height of the aircraft derived from measurements acquired during the flight, a specific energy of the aircraft during an approach phase of the flight. The method further includes determining a reference specific energy of the aircraft, defining an energy range relative to the reference specific energy, and visually displaying, using a first flight instrument of the aircraft, an indicator of the specific energy relative to the energy range.

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Background/Summary

FIELD

[0001] Aspects of the present disclosure relate to aircraft systems, and more specifically, techniques for identifying an unstable approach of an aircraft during flight, based on calculations of a specific energy of the aircraft.

BACKGROUND

[0002] When landing an aircraft, over-energy or low-energy conditions can occur when the flight crew does not properly manage aircraft altitude and/or speed through the approach phase of the flight. These factors result in unstable approaches of the aircraft, which correspond to an increased risk of runway excursions, tail strikes, hard landings, or runway undershoots. According to data from the International Air Transport Association (IATA), unstable approaches contributed to 26% of approach and landing accidents occurring between 2016 and 2020.

[0003] Most cases of unstable approaches result from excessive altitude and/or speed along a final approach segment of the approach phase (e.g., from the final approach fix to touchdown), even though the configuration of landing gear and landing flaps is technically correct. Typically, the pilot in command is responsible to identify that the aircraft is not meeting one or more criteria for a stable approach, and should perform a go-around and/or a missed approach procedure. The Flight Safety Foundation identifies the failure to identify the conditions for stabilized approach as one of the major causes of runway excursions.

SUMMARY

[0004] The present disclosure provides a method of monitoring for an unstable approach of an aircraft during flight in one aspect, the method including: determining, using values of a ground speed and a height of the aircraft derived from measurements acquired during the flight, a specific energy of the aircraft during an approach phase of the flight. The method further includes determining a reference specific energy of the aircraft, defining an energy range relative to the reference specific energy, and visually displaying, using a first flight instrument of the aircraft, an indicator of the specific energy relative to the energy range.

[0005] In one aspect, in combination with any example method above or below, the first flight instrument comprises a primary flight display (PFD), the indicator of the specific energy comprises a graphical element overlaid on a vertical crossbar of a flight director (FD) of the PFD, and one or both of a position and a color of the graphical element is based on the specific energy.

[0006] In one aspect, in combination with any example method above or below, the method further includes visually displaying, using a second flight instrument of the aircraft, an indicator of an actual indicated airspeed relative to a reference speed according to a selected landing flap setting.

[0007] In one aspect, in combination with any example method above or below, the second flight instrument comprises a primary flight display (PFD), the indicator of the actual indicated airspeed comprises an airspeed readout of an airspeed indicator of the PFD, and a color of the airspeed readout is based on the actual indicated airspeed.

[0008] In one aspect, in combination with any example method above or below, the method further includes visually displaying, using a third flight instrument of the aircraft, an indicator of a vertical speed.

[0009] In one aspect, in combination with any example method above or below, the third flight instrument comprises a primary flight display (PFD), the indicator of the vertical speed comprises one or both of a vertical speed readout and a vertical speed pointer of a vertical speed indicator of the PFD, and a color of the indicator is based on the vertical speed.

[0010] In one aspect, in combination with any example method above or below, the method further includes: comparing the specific energy with one or more limits of the energy range, and

presenting, when the specific energy exceeds a limit of the energy range, an audiovisual warning using a fourth flight instrument of the aircraft.

[0011] In one aspect, in combination with any example method above or below, the audiovisual warning comprises a textual warning displayed on a scratchpad of the fourth flight instrument.

[0012] In one aspect, in combination with any example method above or below, the reference specific energy is a sum of a reference ground speed term, and a reference height term based on a threshold elevation.

[0013] The present disclosure provides a method of identifying an unstable approach of an aircraft, performed using a flight management system (FMS) of the aircraft, in one aspect, the method including: determining a reference specific energy of the aircraft. Determining the reference specific energy of the aircraft comprises: determining a reference ground speed term comprising a reference speed according to a selected landing flap setting; and determining a reference height term based in part on data retrieved from a navigation database of the FMS. The method further includes defining an energy range relative to the reference specific energy, and determining, using an actual height and an actual ground speed of the aircraft, a specific energy of the aircraft during an approach phase of the flight. The method further includes generating a signal to cause an indicator of the specific energy of the aircraft, relative to the energy range, to be visually displayed using a flight instrument of the aircraft.

[0014] In one aspect, in combination with any example method above or below, the reference specific energy comprises a sum of the reference ground speed term and the reference height term.

[0015] In one aspect, in combination with any example method above or below, the reference ground speed term comprises a sum of a reference speed according to a selected landing flap setting, a headwind term, and a speed adjustment term.

[0016] In one aspect, in combination with any example method above or below, the reference height term comprises one of the following: when a distance of the aircraft to a runway threshold is between a predetermined distance value and a final approach fix (FAF), a product of the distance and a tangent of an approach phase path angle, plus an altitude of the FAF, minus a threshold elevation, and when the distance is between the FAF and a point that is a predetermined height value above the runway threshold, a product of the distance and a tangent of a final approach path angle, minus the threshold elevation.

[0017] In one aspect, in combination with any example method above or below, defining the energy range comprises: determining a minimum limit as a product of a first gain value and the reference specific energy, and determining a maximum limit as a product of a second gain value and the reference specific energy.

[0018] In one aspect, in combination with any example method above or below, defining the energy range further comprises: receiving one or more user inputs for the first gain value and the second gain value, and determining whether the first gain value and the second gain value are within a permissible range that is based on a margin value.

[0019] In one aspect, in combination with any example method above or below, defining the energy range further comprises: determining a first intermediate limit between the reference specific energy and the minimum limit, and determining a second intermediate limit between the reference specific energy and the maximum limit. The method further includes generating a first signal to cause a flight instrument of the aircraft to present a first audiovisual warning responsive to the specific energy of the aircraft exceeding the first intermediate limit or the second intermediate limit, and generating a second signal to cause the flight instrument to present a second audiovisual warning responsive to the specific energy of the aircraft exceeding the minimum limit or the maximum limit.

[0020] The present disclosure provides a system in one aspect, the system including: one or more computer processors, and a memory storing instructions that when executed by the processor enable performance of an operation of identifying an unstable approach of an aircraft. The

operation includes determining a reference specific energy of the aircraft. Determining the reference specific energy of the aircraft comprises: determining a reference ground speed term comprising a reference speed according to a selected landing flap setting, and determining a reference height term based in part on data retrieved from a navigation database. The operation further includes defining an energy range relative to the reference specific energy, and determining, using an actual height and an actual ground speed of the aircraft, a specific energy of the aircraft during an approach phase. The operation further includes generating a signal to cause an indicator of the specific energy of the aircraft, relative to the energy range, to be visually displayed using a flight instrument of the aircraft.

[0021] In one aspect, in combination with any example system above or below, the reference specific energy comprises a sum of the reference ground speed term and the reference height term.

[0022] In one aspect, in combination with any example system above or below, defining the energy range comprises: determining a minimum limit as a product of a first gain value and the reference specific energy, and determining a maximum limit as a product of a second gain value and the reference specific energy.

[0023] In one aspect, in combination with any example system above or below, the operation further includes: comparing the specific energy with one or more limits of the energy range, and presenting, when the specific energy exceeds a limit of the energy range, an audiovisual warning using a second flight instrument of the aircraft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] So that the manner in which the above recited features can be understood in detail, a more particular description, briefly summarized above, may be had by reference to example aspects, some of which are illustrated in the appended drawings.

[0025] FIG. 1 depicts an exemplary approach and landing of an aircraft, according to one or more aspects.

[0026] FIG. 2 is a block diagram of an example system capable of identifying an unstable approach of an aircraft, according to one or more aspects.

[0027] FIG. 3 is a method of identifying an unstable approach of an aircraft, according to one or more aspects.

[0028] FIG. 4 depicts an exemplary energy range defined relative to a reference specific energy, according to one or more aspects.

[0029] FIG. 5 depicts an exemplary energy range having intermediate limits and a reference specific energy based on landing flap settings, according to one or more aspects.

[0030] FIG. 6 depicts an exemplary Primary Flight Display (PFD) with an indicator of a specific energy relative to the energy range, according to one or more aspects.

[0031] FIG. 7 depicts an exemplary Multipurpose Control and Display Unit (MCDU) with a textual warning displayed on a scratchpad, according to one or more aspects.

DETAILED DESCRIPTION

[0032] The present disclosure provides techniques for identifying an unstable approach of an aircraft during flight, based on calculations of a specific energy of the aircraft during an approach segment of the flight. The specific energy is compared with an energy range defined relative to a reference specific energy, and an indicator of the comparison is communicated using one or more flight instruments. In some aspects, the specific energy is compared with limits of the energy range, which may include intermediate limits, and an audiovisual warning is presented when the specific energy exceeds one of the limits.

[0033] The techniques described herein can improve the situational awareness of the flight crew,

and especially so during the approach segment, rapidly identifying conditions causing an unstable approach of the aircraft. In some aspects, the indicator may be implemented to further provide recommendations to the pilot as to changes in thrust application and/or attitude that will maintain and/or return the aircraft to a stabilized approach (and a balanced energy state). The improved situational awareness contributes to improved flight safety, and also reduces operational costs by mitigating the number of go-around events.

[0034] Conventional solutions that monitor (and alert) for unstable approach rely on sophisticated algorithms that evaluate several combinations of flight parameters and flight envelopes, but these solutions do not consider a total specific energy of the aircraft during the approach. Additionally, these conventional solutions require the adaptation or custom configuration of various avionics systems, which increases their cost. The techniques described herein provide a simple, cost-effective solution that may be implemented using standard avionics systems. For example, in some aspects the calculations of the specific energy are performed by the Flight Management System (FMS) using preexisting parameters, the indicator is displayed using a Primary Flight Display (PFD), and warning messages are visually displayed on PFDs and/or FMS (e.g., Multipurpose Control and Display Unit (MCDU)) scratchpads.

[0035] In the current disclosure, reference is made to various aspects. However, it should be understood that the present disclosure is not limited to specific described aspects. Instead, any combination of the following features and elements, whether related to different aspects or not, is contemplated to implement and practice the teachings provided herein. Additionally, when elements of the aspects are described in the form of “at least one of A and B,” it will be understood that aspects including element A exclusively, including element B exclusively, and including element A and B are each contemplated. Furthermore, although some aspects may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given aspect is not limiting of the present disclosure. Thus, the aspects, features, aspects and advantages disclosed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the invention” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

[0036] Turn now to FIG. 1, which depicts an exemplary approach and landing of an aircraft **105**, according to one or more aspects. In diagram **100**, the aircraft **105** is operated along a predetermined flight path **110** until the aircraft **105** has landed on a runway **115**. The flight path **110** includes a number of phases, including a cruise phase **120** where the aircraft **105** maintains a cruise altitude, a descent phase and an approach phase (shown here for simplicity as a single approach phase **125**) where the aircraft **105** gradually reduces altitude, and a landing phase **130** where the aircraft **105** touches down on the runway **115**. The various phases are not drawn to scale in the diagram **100**.

[0037] One or more of the phases of the flight path **110** may define a plurality of segments. For example, the approach phase **125** may include an initial approach segment, an intermediate approach segment, a final approach segment, and a missed approach segment. The beginning and end of each of the phases (and each of the segments) are defined by particular operational criteria, which will be understood by the person of ordinary skill in the art.

[0038] In various aspects described herein, a specific energy of the aircraft **105** is calculated during the approach phase **125**, and is used to identify whether the aircraft **105** is in an unstable approach. The specific energy is compared with an energy range defined relative to a reference specific energy, and an indicator of the comparison is communicated using one or more flight instruments. Through communicating the indicator, the situational awareness of the flight crew is improved during the approach phase **125**.

[0039] FIG. 2 is a block diagram of an example system **200** capable of identifying an unstable

approach of an aircraft, according to one or more aspects. The features of the system **200** may be used in conjunction with other aspects.

[0040] In the system **200**, the aircraft **105** comprises a FMS **205**, a plurality of sensors **235**, and a plurality of flight instruments **240**. The FMS **205** is an electronic device, disposed onboard the aircraft **105**, that provides navigation, performance, and operations functionalities for the aircraft **105**. As used herein, an “electronic device” generally refers to any device having electronic circuitry that provides a processing or computing capability, and that implements logic and/or executes program code to perform various operations that collectively define the functionality of the electronic device. The functionality of the electronic device includes a communicative capability with one or more other electronic devices, e.g., when connected to a same network. An electronic device may be implemented with any suitable form factor, whether relatively static in nature (e.g., mainframe, computer terminal, server, kiosk, workstation) or mobile (e.g., laptop computer, tablet, handheld, smart phone, wearable device). The communicative capability between electronic devices may be achieved using any suitable techniques, such as conductive cabling, wireless transmission, optical transmission, and so forth.

[0041] As shown, the FMS **205** is coupled with the plurality of sensors **235** and with the plurality of flight instruments **240**. In other aspects, one or more of the flight instruments **240** are integrated into the FMS **205**. The FMS **205** may be coupled with one or more electronic devices that are external to the aircraft **105** through a network **255** (e.g., one or more local area networks (LANs) and/or a wide area network (WAN)).

[0042] Although described as being performed by a single electronic device, in other aspects, the functionalities of the FMS **205** may be performed by a plurality of electronic devices. In one non-limiting example, the FMS **205** comprises a Flight Management Computer (FMC) that allows the flight path to be pre-programmed; an Automatic Flight Control System (AFCS) or Automatic Flight Guidance System (AFGS) that receives information from one or more of the sensors **235** and provides signals for control surfaces of the aircraft **105** and/or recommendations to the pilot; an Aircraft Navigation System that calculates positioning of the aircraft **105** using an inertial reference system, Global Positioning System (GPS), and/or ground-based aids; and an Electronic Flight Instrument System (EFIS) or electromechanical instruments.

[0043] The FMS **205** comprises one or more processors **210** and a memory **215**. The one or more processors **210** are any electronic circuitry, including, but not limited to one or a combination of microprocessors, microcontrollers, application-specific integrated circuits (ASIC), application-specific instruction set processors (ASIP), and/or state machines, that is communicatively coupled to the memory **215** and controls the operation of the FMS **205**. The one or more processors **210** are not limited to a single processing device and may encompass multiple processing devices.

[0044] The one or more processors **210** may include other hardware that operates software to control and process information. In some aspects, the one or more processors **210** execute software stored in the memory **215** to perform any of the functions described herein. The one or more processors **210** control the operation and administration of the FMS **205** by processing information (e.g., information received from input devices and/or communicatively coupled electronic devices).

[0045] The memory **215** may store, either permanently or temporarily, data, operational software, or other information for the one or more processors **210**. The memory **215** may include any one or a combination of volatile or non-volatile local or remote devices suitable for storing information. For example, the memory **215** may include random access memory (RAM), read only memory (ROM), magnetic storage devices, optical storage devices, or any other suitable information storage device or a combination of these devices. The software represents any suitable set of instructions, logic, or code embodied in a computer-readable storage medium. For example, the software may be embodied in the memory **215**, a disk, a CD, or a flash drive. In particular embodiments, the software may include an application executable by the one or more processors **210** to perform one or more of the functions described herein.

[0046] In this example, the memory **215** stores an approach monitoring service **220** that acquires measurements from the plurality of sensors **235** (or values of one or more parameters derived from the measurements), and determines a specific energy **225** of the aircraft **105** during the approach phase **125** of the flight. The types of the plurality of sensors **235** used in the aircraft **105** will be known to the person of ordinary skill in the art. In some aspects, the approach monitoring service **220** calculates the actual total specific energy of the aircraft **105** throughout the approach phase **125** (e.g., at any ground position between the point at which the initial approach flap (also referred to herein as a landing flap) setting is selected, and **50** feet above the threshold of the runway **115**). In some aspects, the approach monitoring service **220** compares the specific energy with an energy range, and generates a signal to cause an indicator of the specific energy, relative to the energy range, to be visually displayed using a flight instrument of the aircraft **105**. The energy range is defined relative to a reference specific energy of the aircraft **105**, which in some aspects is also calculated by the approach monitoring service **220**. Further discussion of calculation of the energy range is provided below with respect to FIGS. **4** and **5**.

[0047] The indicator is displayed using one of the plurality of flight instruments **240**. As shown, the flight instruments **240** comprise a PFD **245** and an MCDU **250**, although other numbers and types of flight instruments **240** are also contemplated. The flight instruments **240** used in the aircraft **105** will be known to the person of ordinary skill in the art. In some aspects, the indicator is displayed using a PFD **245** of the aircraft **105**. In some alternate aspects, the indicator may be displayed using another display of the aircraft **105**, such as a multi-function display (MFD). Generally, the PFD **245** comprises an electronic display implementing any suitable display technology (e.g., CRT, (O)LED, LCD). The PFD **245** displays various flight information, such as an airspeed indicator, a turn coordinator, an attitude indicator, a heading indicator, an altimeter, and a vertical speed indicator. Although the PFD **245** may display the flight information in any suitable manner, in some aspects, the PFD **245** displays representations of mechanical flight instruments (e.g., a cluster of analog gauges). In one example implementation, the indicator of the specific energy comprises a graphical element overlaid on a vertical flight director (F/D) bar of the PFD **245**. Further discussion of displaying the indicator using the PFD **245** is provided below with respect to FIG. **6**.

[0048] In some aspects, the approach monitoring service **220** compares the specific energy with one or more limits of the energy range, and generates signals to cause a flight instrument **240** of the aircraft **105** to present audiovisual warning(s) responsive to the specific energy of the aircraft exceeding the limit(s). In one example implementation, the audiovisual warning(s) comprise textual warnings that are displayed on a scratchpad of a flight instrument **240**, such as the PFD **245** or the MCDU **250**. Further discussion of presenting the audiovisual warning(s) is provided below with respect to FIG. **7**.

[0049] In some aspects, the memory **215** further stores a navigation database **230** containing information that is used to construct a flight plan for the aircraft **105**. For example, the navigation database **230** may include some or all of the following information: waypoints/intersections, airways, radio navigation aids, airports, runways, and/or standard procedures (SID, STAR, IAP, holds). The information may be stored in the navigation database **230** with any suitable formatting.

[0050] In some alternate aspects, the approach monitoring service **220** may be implemented using one or more electronic devices other than the FMS **205**, for example, an electronic device that is wirelessly coupled with the FMS **205** through the network **255**. Such an electronic device may be located onboard the aircraft **105** (e.g., another electronic device of the aircraft **105**, a portable electronic device) or external to the aircraft **105**.

[0051] FIG. **3** is a method **300** of identifying an unstable approach of an aircraft, according to one or more aspects. The method **300** may be used in conjunction with other aspects. For example, in some aspects the method **300** may be performed by the approach monitoring service **220** and/or other components of the FMS **205**. In some aspects, the method **300** may be performed, partly or fully, by electronic device(s) that are separate from the FMS **205**.

[0052] The method **300** begins at block **305**, where the FMS **205** acquires measurements during flight. In some aspects, the measurements include positional information for the aircraft **105**, such as samples of a GPS signal, an Inertial Navigation System signal, a radio navigation signal, and so forth. Other measurements are also contemplated.

[0053] At block **310**, the FMS **205** determines, using values of a ground speed and a height of the aircraft **105**, a specific energy of the aircraft **105** during an approach phase **125** of the flight. In some aspects, the specific energy of the aircraft **105** is determined according to Equation (1):

$$[00001] e_{\text{actual}} = \frac{GS^2}{2} + g \cdot \text{Math. } H \quad (1)$$

where GS represents the actual ground speed of the aircraft **105**, H represents the actual height of the aircraft **105** above a threshold elevation, and g represents the gravitational acceleration constant. In some aspects, GS is computed by the FMS **205** or by an Air Data Inertial Reference System (ADIRS) coupled with the FMS **205**. In some aspects, H is computed by the FMS **205**.

[0054] In some aspects, the FMS **205** determines values of the specific energy at points of the flight path **110**, beginning from where the approach flap is first deployed and ending at **50** feet above the threshold of the runway **115** (i.e., above ground level (AGL)). The FMS **205** may determine the values of the specific energy with any suitable sampling period, such as one (1) second.

[0055] At block **315**, the FMS **205** determines a reference specific energy of the aircraft **105**. In some aspects, determining the reference specific energy of the aircraft **105** comprises determining a reference ground speed term (block **320**) and determining a reference height term (block **325**), and the reference specific energy comprises a sum of the reference ground speed term and the reference height term. In some aspects, the reference specific energy is determined according to Equation (2):

$$[00002] e_{\text{ref}} = \frac{GS_{\text{ref}}^2}{2} + g \cdot \text{Math. } H_{\text{ref}} \quad (2)$$

where

$$[00003] \frac{GS_{\text{ref}}^2}{2}$$

represents the reference ground speed term and g. Href represents the reference height term.

[0056] In some aspects, the reference ground speed term

$$[00004] \frac{GS_{\text{ref}}^2}{2}$$

comprises a reference speed that is determined according to a selected landing flap setting. In some aspects, the reference ground speed term comprises a sum of the reference speed, a headwind term, and a speed adjustment term. For example, when landing flaps are deployed, the reference ground speed term is determined according to Equation (3):

$$[00005] GS_{\text{ref}} = V_{\text{REF}} + \Delta V + HW \quad (3)$$

and when approach flaps are deployed, the reference ground speed term is determined according to Equation (4):

$$[00006] GS_{\text{ref}} = V_{\text{REF}40} + X + HW \quad (4)$$

where VREF represents the reference speed according to the selected landing flap setting, as determined by the FMS **205**, ΔV represents the speed adjustment term input to the FMS **205** by the pilot for the approach (e.g., with a minimum value of 5 kt), and HW represents the headwind term computed by the FMS **205** based on pilot input (e.g., reported by air traffic control). VREF40 represents the reference speed for a 40° flap setting, and X represents a pilot-selectable speed adjustment term (e.g., default five (5) kt). Thus, for a zero wind condition the minimum selectable speed would be (VREF+5 kt).

[0057] In some aspects, the reference height term g. Href is based in part on data retrieved from the navigation database **230** of the FMS **205**. In some aspects, Href is a reference height above the threshold elevation, and is determined according to Equations (5) and (6):

$$[00007] H_{\text{ref}} = D \cdot \text{Math. } \tan(\text{GPA}) - \text{Elev}_{\text{thr}} \quad (5)$$

when the distance D to the threshold is between a final approach fix (FAF) and a point that is a predetermined height value (e.g., 50 feet) above the threshold, and

[00008] $H_{ref} = D \cdot \tan(\text{Path_angle}) + \text{ALT}_{\text{FAF}} - \text{Elev_thr}$ (6)

[0058] when the distance D to the threshold is between a predetermined distance value (e.g., 15 nautical miles) and the FAF. Different values of the predetermined height value and the predetermined distance values are also contemplated.

[0059] Thus, according to Equation (5), the FMS **205** determines the reference height Href as a product of the distance D and a tangent of a final approach path angle GPA, minus the threshold elevation Elev_thr. According to Equation (6), the FMS **205** determines the reference height Href as a product of the distance D and a tangent of an approach segment path angle Path_angle, plus an altitude of the FAF ALT.sub.FAF, minus the threshold elevation Elev_thr. In some aspects, the FMS **205** computes the distance D and the approach segment path angle Path_angle, and acquires the final approach path angle GPA and the landing threshold elevation Elev_thr from the navigation database **230**.

[0060] At block **330**, the FMS **205** defines an energy range relative to the reference specific energy. In some aspects, defining the energy range comprises receiving one or more user inputs for a first gain value Kmin and a second gain value Kmax (block **335**). For example, the first gain value Kmin and the second gain value Kmax may be configurable in the FMS **205** via a dedicated page (e.g., an extension of a Performance Factors page).

[0061] At block **340**, the FMS **205** determines whether the first gain value Kmin and the second gain value Kmax are within a permissible range that is based on a margin value. In some aspects, the permissible range is defined according to Equations (7) and (8):

[00009] $1 - \text{Margin} < K_{min} < 0.99$ (7) $1.01 < K_{max} < 1 + \text{Margin}$ (8)

In some aspects, a default value of the margin is 0.2. However, other margin values and other parameters of the permissible range are also contemplated.

[0062] Assuming that the first gain value Kmin and the second gain value Kmax are within the permissible range, the method **300** proceeds to block **345**, where the FMS **205** determines a minimum limit of the energy range as a product of the first gain value Kmin and the reference specific energy e_ref, as represented in Equation (9):

[00010] $e_{min} = K_{min} * e_{ref}$ (9)

At block **350**, the FMS **205** determines a maximum limit of the energy range as a product of a second gain value Kmax and the reference specific energy e_ref, as represented in Equation (10):

[00011] $e_{max} = K_{max} * e_{ref}$ (10)

[0063] Turn now to FIG. 4, which depicts an exemplary energy range **420** defined relative to the reference specific energy e_ref, according to one or more aspects. In graph **400**, plot **405** represents the reference specific energy e_ref for the approach phase. Plot **410** represents the minimum limit e_min, and plot **415** represents the maximum limit e_max. Each of the plots **405**, **410**, **415** decreases proportionally with a decrease in the distance to touchdown of the aircraft **105**. For example, the reference ground speed term GSref of the aircraft **105** is decreased as the aircraft **105** nears the runway (i.e., as the distance to touchdown decreases). As shown in Equations (2), (9), and (10), the reference specific energy e_ref, the minimum limit e_min, and the maximum limit e_max each decrease according to the square of the decrease in the reference ground speed term GSref.

[0064] The point **425** represents a single measurement of a specific energy e_actual of the aircraft **105**, e.g., as calculated by the FMS **205**. The point **425** falls within the energy range **420**, indicating that the specific energy of the aircraft **105** is not causing an unstable approach. As will be discussed in greater detail below, in some aspects no audiovisual warning (or other alert) will be generated by the FMS **205** for this condition.

[0065] In some aspects, one or more intermediate limits may be defined in the energy range **420** between the minimum limit and the maximum limit. Such intermediate limit(s) may be used to trigger different alerts and/or warnings to improve the situational awareness of the pilot. Returning to FIG. 3, at block **355** the FMS **205** determines a first intermediate limit between the reference

specific energy and the minimum limit, and at block **360** the FMS **205** determines a second intermediate limit between the reference specific energy and the maximum limit.

[0066] FIG. **5** depicts an exemplary energy range having intermediate limits and a reference specific energy based on landing flap settings, according to one or more aspects. In graph **500**, the plot **405** again represents the reference specific energy e_{ref} for the approach phase. However, instead of a single, linear plot as in the graph **400**, the plot **405** includes a plurality of segments corresponding to a plurality of landing flap settings **505**, **510**, **515**, **520** according to a flap extension schedule for the approach phase. In one non-limiting example, the landing flap setting **505** is “Flap 1”, the landing flap setting **510** is “Flap 5”, the landing flap setting **515** is “Flap 15”, and the landing flap setting **520** is “Flap 40”, although other values and sequences of landing flap settings are also contemplated. As shown, the plot **405** includes relatively steep segments where the landing flaps transition from one setting to the next, and flatter segments between the transitions.

[0067] The plots **410**, **415** define the minimum limit and the maximum limit of the energy range **420**. Like the plot **405**, the plots **410**, **415** each include different segments that correspond to the landing flap settings **505**, **510**, **515**, **520**. The graph **500** further includes a plot **525** representing a first intermediate limit between the reference specific energy (the plot **405**) and the minimum limit (the plot **410**), and a plot **530** representing a second intermediate limit between the reference specific energy (the plot **405**) and the maximum limit (the plot **415**). The plots **525**, **530** each include different segments that correspond to the landing flap settings **505**, **510**, **515**, **520**. The first intermediate limit and the second intermediate limit may have any suitable values, which may correspond to particular values of the specific energy or relative to the reference specific energy, the minimum limit, and/or the maximum limit (e.g., the midpoint between the reference specific energy and the maximum limit).

[0068] As mentioned above, the intermediate limits represented by the plots **525**, **530** may be used to provide alerts and/or warnings with greater resolution to the pilot of the aircraft **105**. For values of a specific energy of the aircraft **105** falling between the plots **525**, **530**, the specific energy of the aircraft **105** is not causing (nor particularly likely to cause) an unstable approach. In some aspects, no audiovisual warning (or other alert) will be generated by the FMS **205** for this condition.

[0069] In some aspects, for values of the specific energy between the minimum limit (the plot **410**) and the first intermediate limit (the plot **525**), the FMS **205** generates a first audiovisual warning **540** (“LOW ENERGY”). In some aspects, for values of the specific energy less than the minimum limit, the FMS **205** generates a second audiovisual warning **535** (“LOW ENERGY/GO-AROUND”) directing the pilot to perform a go-around.

[0070] In some aspects, for values of the specific energy between the second intermediate limit (the plot **530**) and the maximum limit (the plot **415**), the FMS **205** generates a third audiovisual warning **545** (“HIGH ENERGY”). As shown, the point **425** falls between the second intermediate limit and the maximum limit. In some aspects, for values of the specific energy greater than the maximum limit, the FMS **205** generates a fourth audiovisual warning **550** (“HIGH ENERGY/GO-AROUND”) directing the pilot to perform a go-around.

[0071] The text of the various audiovisual warnings **535**, **540**, **545**, **550** has been provided as one non-limiting example. Other texts, visual elements, and/or audio elements are also contemplated. Further, although a first intermediate limit and a second intermediate limit are described, other aspects may include different numbers of intermediate limits that provide a different level of granularity to the pilot.

[0072] Although described in terms of the FMS **205** defining the energy range **420**, in some aspects, features of the energy range **420** may be displayed in the aircraft **105** (e.g., in a Vertical Situation Display (VSD)) to improve the situational awareness of the pilot.

[0073] Returning to FIG. **3**, at block **365** the FMS **205** visually displays, using a flight instrument **240** of the aircraft **105**, an indicator of the specific energy relative to the energy range. In some aspects, the FMS **205** generates a signal causing the indicator to be visually displayed. In some

aspects, the flight instrument **240** comprises a PFD **245** of the aircraft **105**.

[0074] FIG. **6** depicts an exemplary PFD **245** with the indicator, according to one or more aspects. In diagram **600**, the PFD **245** displays (from left to right) an airspeed indicator **605**, an attitude indicator **610**, an altimeter **615**, and a vertical speed indicator **620**. A flight director (FD) **640** comprising a dual crossbar is overlaid on the attitude indicator **610**. An indicator **630-1** of the specific energy relative to the energy range comprises a graphical element overlaid on a vertical crossbar **645** of the FD **640**.

[0075] As shown, the indicator **630-1** is a green circle overlaid on the center of the vertical crossbar **645**. In some aspects, one or more properties of the indicator **630-1**, such as the position, color, shape, and size of the indicator **630-1**, are controlled based on the specific energy.

[0076] In some aspects, the specific energy controls one or both of a position and a color of the indicator **630-1**. In some aspects, the range of the indicator **630-1** along the vertical crossbar **645** is determined according to Equation (11):

$$[00012] \ L = K \cdot \text{Math.} \frac{e_{\text{actual}} - e_{\text{ref}}}{e_{\text{ref}}} \quad (11)$$

where K represents a scale factor to fit the range onto a length of vertical crossbar **645**.

[0077] In some aspects, where the specific energy is between the first intermediate limit and the second intermediate limit, the indicator **630-1** is green (or white) and positioned on the center of the vertical crossbar **645**. Where the specific energy is between the second intermediate limit and the maximum limit, the indicator is amber and positioned above the center of the vertical crossbar **645** (shown as outline **630-2**). Where the specific energy exceeds the maximum limit, the indicator is red and positioned at the top of the vertical crossbar **645** (shown as outline **630-3**).

[0078] Continuing the example, where the specific energy is between the first intermediate limit and the minimum limit, the indicator is amber and positioned below the center of the vertical crossbar **645** (shown as outline **630-4**). Where the specific energy exceeds the minimum limit, the indicator is red and positioned at the bottom of the vertical crossbar **645** (shown as outline **630-5**).

[0079] At block **370**, the FMS **205** compares the specific energy with one or more limits of the energy range. In some aspects, the one or more limits include the minimum limit, the maximum limit, the first intermediate limit, and the second intermediate limit.

[0080] At block **375**, the FMS **205** presents, when the specific energy exceeds a limit of the energy range, an audiovisual warning using a flight instrument **240** of the aircraft **105**. As used herein, an “audiovisual warning” may be presented audibly, visually, or both. In some aspects, the audiovisual warning is a textual warning displayed on a scratchpad of the FMS **205** or the MCDU **250**.

[0081] In some aspects, the audiovisual warning may further include an audible warning corresponding to the textual warning. The audible warning may be presented each time the infringing condition is met. In some aspects, the audible warning may be latched on defined intervals, e.g., depending on radio altitudes. For example, between the FAF altitude and 1000 ft AGL, the audible warning may be triggered every five (5) seconds. Between 1000 ft and 500 ft AGL, the audible warning may be triggered every three (3) seconds, indicating a greater urgency. In some aspects, the audible warnings may be inhibited by the operator below a certain radio altitude (e.g., 500 ft AGL).

[0082] Where the specific energy exceeds the second intermediate limit (but is less than the maximum limit), the FMS **205** generates the third audiovisual warning **545** (“HIGH ENERGY”). Where the specific energy exceeds the maximum limit, the FMS **205** generates the fourth audiovisual warning **550** (“HIGH ENERGY/GO-AROUND”). Where the specific energy exceeds the first intermediate limit (but is greater than the minimum limit), the FMS **205** generates the first audiovisual warning **540** (“LOW ENERGY”). Where the specific energy exceeds the minimum limit, the FMS **205** generates the second audiovisual warning **535** (“LOW ENERGY/GO-AROUND”).

[0083] FIG. **7** depicts an exemplary MCDU **250** with a textual warning displayed on a scratchpad,

according to one or more aspects. In diagram **700**, the MCDU **250** comprises a display area **705** arranged between a first row of virtual buttons **710-1** and a second row of virtual buttons **710-2**. The scratchpad is displayed on the display area **705**, and the third audiovisual warning **545** is depicted at location **715** of the scratchpad.

[0084] At block **380**, the FMS **205** visually displays, using a flight instrument **240** of the aircraft **105**, an indicator of actual indicated airspeed relative to reference speed according to selected landing flap setting. In some aspects, the flight instrument **240** comprises the PFD **245**, and the indicator of the actual indicated airspeed comprises an airspeed readout **625** of the airspeed indicator **605** of the PFD. In some aspects, a color of the airspeed readout **625** is based on the actual indicated airspeed.

[0085] In some aspects, where the actual indicated airspeed is between V_{ref} and $V_{ref}+5$ knots indicated airspeed (KIAS), the color of the airspeed readout **625** is white (or green). Where the actual indicated airspeed is between $V_{ref}+5$ and $V_{ref}+X$ KIAS, or is between V_{ref} and $V_{ref}-X$ KIAS, the color of the airspeed readout **625** is amber. Where the actual indicated airspeed is greater than $V_{ref}+X$ KIAS, or less than $V_{ref}-X$ KIAS, the color of the airspeed readout **625** is red. Here, X represents an operator-defined limit for stable approach criteria (e.g., loaded by the operator into the FMS Configuration DB).

[0086] At block **385**, the FMS **205** visually displays, using a flight instrument **240** of the aircraft **105**, an indicator of a vertical speed. In some aspects, the flight instrument **240** comprises the PFD **245**, and the indicator of the vertical speed comprises one or both of a vertical speed readout and a vertical speed pointer **635-1** of the vertical speed indicator **620**. The color of the indicator is based on the vertical speed.

[0087] In some aspects, where the vertical speed is between first and second threshold values (e.g., +500 FT/Min), the vertical speed pointer **635-1** and/or the vertical speed readout are white (or green). Where the vertical speed is between the first threshold value (e.g., +500 FT/Min) and a third threshold value (e.g., +1000 FT/Min) (as shown by outline **635-4**), or between the second threshold value (e.g., -500 FT/Min) and a fourth threshold value (e.g., -1000 FT/Min) (as shown by outline **635-2**), the vertical speed pointer **635-1** and/or the vertical speed readout are amber. Where the vertical speed is greater than the third threshold value (e.g., +1000 FT/Min) (as shown by outline **635-5**), or less than the fourth threshold value (e.g., -1000 FT/Min) (as shown by outline **635-3**), the vertical speed pointer **635-1** and/or the vertical speed readout are red. The method **300** ends following completion of block **385**.

[0088] As will be appreciated by one skilled in the art, aspects described herein may be embodied as a system, method or computer program product. Accordingly, aspects may take the form of an entirely hardware aspect, an entirely software aspect (including firmware, resident software, micro-code, etc.) or an aspect combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects described herein may take the form of a computer program product embodied in one or more computer readable storage medium(s) having computer readable program code embodied thereon.

[0089] Program code embodied on a computer readable storage medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0090] Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local

area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0091] Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems), and computer program products according to aspects of the present disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0092] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0093] The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process such that the instructions which execute on the computer, other programmable data processing apparatus, or other device provide processes for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0094] The flowchart illustrations and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various aspects of the present disclosure. In this regard, each block in the flowchart illustrations or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order or out of order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0095] While the foregoing is directed to aspects of the present disclosure, other and further aspects of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of monitoring for an unstable approach of an aircraft during flight, the method comprising: determining, using values of a ground speed and a height of the aircraft derived from measurements acquired during the flight, a specific energy of the aircraft during an approach phase of the flight; determining a reference specific energy of the aircraft; defining an energy range relative to the reference specific energy; and visually displaying, using a first flight instrument of the aircraft, an indicator of the specific energy relative to the energy range.
2. The method of claim 1, wherein the first flight instrument comprises a primary flight display (PFD), wherein the indicator of the specific energy comprises a graphical element overlaid on a

vertical crossbar of a flight director (FD) of the PFD, and wherein one or both of a position and a color of the graphical element is based on the specific energy.

3. The method of claim 1, further comprising: visually displaying, using a second flight instrument of the aircraft, an indicator of an actual indicated airspeed relative to a reference speed according to a selected landing flap setting.

4. The method of claim 3, wherein the second flight instrument comprises a primary flight display (PFD), wherein the indicator of the actual indicated airspeed comprises an airspeed readout of an airspeed indicator of the PFD, and wherein a color of the airspeed readout is based on the actual indicated airspeed.

5. The method of claim 1, further comprising: visually displaying, using a third flight instrument of the aircraft, an indicator of a vertical speed.

6. The method of claim 5, wherein the third flight instrument comprises a primary flight display (PFD), wherein the indicator of the vertical speed comprises one or both of a vertical speed readout and a vertical speed pointer of a vertical speed indicator of the PFD, and wherein a color of the indicator is based on the vertical speed.

7. The method of claim 1, further comprising: comparing the specific energy with one or more limits of the energy range; and presenting, when the specific energy exceeds a limit of the energy range, an audiovisual warning using a fourth flight instrument of the aircraft.

8. The method of claim 7, wherein the audiovisual warning comprises a textual warning displayed on a scratchpad of the fourth flight instrument.

9. The method of claim 1, wherein the reference specific energy is a sum of a reference ground speed term, and a reference height term based on a threshold elevation.

10. A method of identifying an unstable approach of an aircraft, performed using a flight management system (FMS) of the aircraft, the method comprising: determining a reference specific energy of the aircraft, wherein determining the reference specific energy of the aircraft comprises: determining a reference ground speed term comprising a reference speed according to a selected landing flap setting; and determining a reference height term based in part on data retrieved from a navigation database of the FMS; defining an energy range relative to the reference specific energy; determining, using an actual height and an actual ground speed of the aircraft, a specific energy of the aircraft during an approach phase of the flight; and generating a signal to cause an indicator of the specific energy of the aircraft, relative to the energy range, to be visually displayed using a flight instrument of the aircraft.

11. The method of claim 10, wherein the reference specific energy comprises a sum of the reference ground speed term and the reference height term.

12. The method of claim 11, wherein the reference ground speed term comprises a sum of a reference speed according to a selected landing flap setting, a headwind term, and a speed adjustment term.

13. The method of claim 11, wherein the reference height term comprises one of the following: when a distance of the aircraft to a runway threshold is between a predetermined distance value and a final approach fix (FAF), a product of the distance and a tangent of an approach phase path angle, plus an altitude of the FAF, minus a threshold elevation, and when the distance is between the FAF and a point that is a predetermined height value above the runway threshold, a product of the distance and a tangent of a final approach path angle, minus the threshold elevation.

14. The method of claim 10, wherein defining the energy range comprises: determining a minimum limit as a product of a first gain value and the reference specific energy; and determining a maximum limit as a product of a second gain value and the reference specific energy.

15. The method of claim 14, wherein defining the energy range further comprises: receiving one or more user inputs for the first gain value and the second gain value; and determining whether the first gain value and the second gain value are within a permissible range that is based on a margin value.

16. The method of claim 14, wherein defining the energy range further comprises: determining a first intermediate limit between the reference specific energy and the minimum limit; and determining a second intermediate limit between the reference specific energy and the maximum limit, and wherein the method further comprises: generating a first signal to cause a flight instrument of the aircraft to present a first audiovisual warning responsive to the specific energy of the aircraft exceeding the first intermediate limit or the second intermediate limit; and generating a second signal to cause the flight instrument to present a second audiovisual warning responsive to the specific energy of the aircraft exceeding the minimum limit or the maximum limit.

17. A system comprising: one or more computer processors; and a memory storing instructions that when executed by the processor enable performance of an operation of identifying an unstable approach of an aircraft, the operation comprising: determining a reference specific energy of the aircraft, wherein determining the reference specific energy of the aircraft comprises: determining a reference ground speed term comprising a reference speed according to a selected landing flap setting; and determining a reference height term based in part on data retrieved from a navigation database; defining an energy range relative to the reference specific energy; determining, using an actual height and an actual ground speed of the aircraft, a specific energy of the aircraft during an approach phase; and generating a signal to cause an indicator of the specific energy of the aircraft, relative to the energy range, to be visually displayed using a flight instrument of the aircraft.

18. The system of claim 17, wherein the reference specific energy comprises a sum of the reference ground speed term and the reference height term.

19. The system of claim 17, wherein defining the energy range comprises: determining a minimum limit as a product of a first gain value and the reference specific energy; and determining a maximum limit as a product of a second gain value and the reference specific energy.

20. The system of claim 17, the operation further comprising: comparing the specific energy with one or more limits of the energy range; and presenting, when the specific energy exceeds a limit of the energy range, an audiovisual warning using a second flight instrument of the aircraft.
